

Kevin Chu

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PhD computational scientist and backend software engineer. 6+ years building production systems where research meets infrastructure. Fluent in both the science and the engineering required to turn ambiguous research needs into systems researchers depend on.

Education and Skills

Georgia Tech PhD Materials Science and Eng.	07/2022
UCLA B.S. Materials Science and Eng. (Computer Science Minor)	06/2017

Languages & Tools: Python (NumPy, SciPy, Pandas), C++17, Fortran, Java; HPC cluster administration (SLURM); AWS/Azure; Docker, CI/CD tooling (Jenkins, GitHub Actions, ArgoCD, Tekton, Atlantis); Terraform; Kubernetes; MPI/parallel programming; gRPC/Protobuf; Iceberg/SQL/DuckDB; Git

Research & Collaboration: Scientific software development, HPC/Distributed systems design, data analysis and visualization (Jupyter), technical writing (design docs, feature specs), cross-functional collaboration, solutions engineering

Relevant Experience

Software Engineer	04/2025 – Present
Rescale San Francisco, CA	

- **Designed and administered** operational visibility infrastructure for compute job orchestration (AWS S3/Iceberg, event streaming), processing 100k+ events/day to surface HPC job lifecycle telemetry and **guide targeted engineering investment**.
- **Championed org-wide adoption** of Apache Iceberg (AWS S3 Tables) from proposal through execution — presentations, workshops, and automated onboarding tooling (Terraform CLI, Agent-skill automation)
- Identified and closed early-access feature gaps in cloud infrastructure (AWS S3 Tables) through direct **engagement with external AWS** engineering stakeholders
- **Triaged platform-wide compute orchestration incidents** as primary 24/7 on-call across multi-cloud distributed environments, conducting rapid root cause analysis to restore compute availability
- Delivered full-service **monitoring and KPI dashboards** (Datadog, Grafana, Superset) across engineering, product, and customer success teams

Software Engineer	01/2024 – 04/2025
Science Corporation Alameda, CA	

- Built a scalable BCI control system processing 50GB+ neural data per session with <20ms real-time ML inference for closed-loop feedback, automated experiment pipelines, and modular architecture
- Optimized electrophysiology data ingestion by three orders of magnitude to support 30kHz raw neural data storage; authored a C++ wrapper for DuckDB's C API to standardize data access across teams
- Owned the full neural engineering data stack — gRPC service definitions, protobuf/database schema design, and client library implementation — delivering regular releases to internal neuroscience and behavior teams

Scientific Software Developer	05/2023 – 01/2024
Mat3ra Walnut Creek, CA	

- Fully containerized a multi-node OpenPBS-based dynamically scaling cloud HPC compute system.
- **Developed a dynamic load analysis tool** to detect and terminate idle compute nodes and fine-tune provisioning parameters based on usage patterns, further reducing cloud expenses.
- **Refactored a monolithic Python backend** into a domain-driven services architecture, boosting scalability and fault tolerance.

Materials Design Engineer

08/2022 – 5/2023

Questek Innovations | Evanston, IL

- **Architected a cloud-based computing workflow** (Docker, AWS ECR, AWS Batch) to automate high-dimensional material fatigue experiments, replacing manual processes and improving efficiency and scalability.
- **Translated cutting-edge materials science research** (physics-based and empirical models) into actionable manufacturing strategies for Fortune 500 and DoD clients, driving real-world engineering impact.
- **Led a \$100K materials design pilot** as technical PM, delivering all milestones on time and on budget, and securing a \$400K follow-on contract.

PhD - Graduate Research Assistant, Teaching Fellowship

08/2017 – 7/2022

Georgia Tech | Atlanta, GA

- Contributed to massively parallelized (MPI), open-source tools - LAMMPS (C++) and <http://www.pycac.org/> (Fortran90)- tailored for execution on HPC clusters (1000+ CPUs).
- **Automated high-throughput computational study workflows** and data curation/analysis to investigate nanoscale material behavior in alloys and semiconductors
- **Bridged computational and experimental studies** of novel materials behavior in collaboration with top national research laboratories (Sandia National Labs, Naval Research Laboratory)